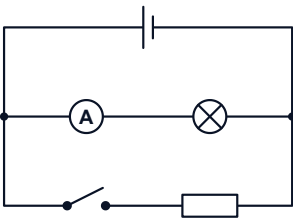


# Series and parallel circuits

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

|     | QUESTION                                                                                                                                                                                            | ANSWER                                                                                                         | ✓                        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------|
| 1   | Name the two ways components can be joined in a circuit.                                                                                                                                            | In series and in parallel                                                                                      | <input type="checkbox"/> |
| 2★  | Describe the current through components in series.                                                                                                                                                  | It is the same through each component.                                                                         | <input type="checkbox"/> |
| 3   | Describe the potential difference across components in series.                                                                                                                                      | The total potential difference is shared between them.                                                         | <input type="checkbox"/> |
| 4   | Describe the total resistance of components connected in series.                                                                                                                                    | It is the sum of the individual resistances.                                                                   | <input type="checkbox"/> |
| 5   | Describe the current through components in parallel.                                                                                                                                                | The total current is the sum of the currents in the branches.                                                  | <input type="checkbox"/> |
| 6   | The lamp in this circuit is lit. Describe what happens to the ammeter reading when the switch is closed.<br><br> | It does not change.<br>The lamp is on a separate branch, so the potential difference across it stays the same. | <input type="checkbox"/> |
| 7★  | Describe the potential difference across components in parallel.                                                                                                                                    | It is the same across each component.                                                                          | <input type="checkbox"/> |
| 8★  | Describe the total resistance of two resistors connected in parallel.                                                                                                                               | It is less than the resistance of the smaller resistor.                                                        | <input type="checkbox"/> |
| 9   | Explain why adding resistors in series increases the total resistance.                                                                                                                              | The potential difference is shared between more resistors, so the current is smaller.                          | <input type="checkbox"/> |
| 10★ | Explain why adding resistors in parallel decreases the total resistance.                                                                                                                            | Each new branch gives the charge another path, so the total current increases.                                 | <input type="checkbox"/> |

| QUESTION |                                                                                        | ANSWER                                                                       | ✓                        |
|----------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------|
| 11       | <b>Lamps are connected in series and one breaks. State what happens to the others.</b> | They all go out, because the circuit is broken.                              | <input type="checkbox"/> |
| 12       | <b>Give two advantages of connecting lamps in parallel rather than in series.</b>      | Each lamp can be switched separately, and if one breaks the others stay lit. | <input type="checkbox"/> |